

HILL AFB, UT, USA

WMO: 725755

Lat: 41.117N

Lon: 111.967W

Elev: 1460

StdP: 84.97

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24101

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.5	-10.6	-17.7	0.9	-9.6	-15.3	1.2	-7.5	12.5	-3.0	11.1	-0.8	4.8	100	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	34.6	16.0	33.2	15.6	32.1	15.4	18.1	29.4	17.4	29.1	16.7	28.7	3.2	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.6	12.4	21.9	12.9	11.1	22.3	11.8	10.3	22.3	56.9	29.2	54.3	29.2	52.0	28.7	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.8	9.2	8.4	DB	-15.4	36.3	2.5	1.4	-17.2	37.3	-18.6	38.1	-20.0	38.9	-21.8	39.9
			WB	-16.1	19.4	2.4	1.4	-17.8	20.4	-19.3	21.2	-20.6	22.0	-22.4	23.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-1.6	0.7	5.9	9.1	14.4	20.1	25.2	24.1	18.4	10.7	4.3	-1.2		
	DBStd	10.16	4.54	4.49	4.49	4.43	4.58	4.51	2.80	2.71	4.45	4.57	4.89	4.70		
	HDD10.0	1423	359	261	143	70	13	1	0	0	3	47	179	348		
	HDD18.3	3273	617	493	387	279	138	34	1	2	52	239	422	607		
	CDD10.0	1746	0	1	14	42	148	303	472	438	254	68	7	0		
	CDD18.3	553	0	0	0	1	15	86	215	182	53	2	0	0		
	CDH23.3	6111	0	0	0	13	171	999	2491	1917	499	21	0	0		
	CDH26.7	2643	0	0	0	1	39	393	1230	839	139	2	0	0		
(14) Wind	WSAvg	4.1	3.6	4.1	4.3	4.3	4.1	4.3	4.3	4.4	4.4	4.1	3.9	3.7		
(15) Precipitation	PrecAvg	435	33	34	43	53	48	33	15	19	39	41	40	36		
	PrecMax	872	80	82	130	143	139	107	48	82	140	107	90	136		
	PrecMin	210	0	2	2	6	0	0	0	1	0	0	1	1		
	PrecStd	127	20	23	31	34	38	31	15	20	37	27	23	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	14.0	21.0	25.0	30.2	35.1	37.1	35.1	31.9	26.0	17.8	12.1		
		MCWB	5.0	5.4	9.3	11.3	14.1	15.9	16.8	16.3	15.6	12.4	8.0	6.1		
	2%	DB	7.7	11.2	17.9	22.1	27.5	32.7	35.2	33.6	30.1	22.7	15.6	9.2		
		MCWB	3.5	4.6	7.8	10.0	13.1	14.7	16.1	15.9	14.8	11.2	7.5	4.5		
	5%	DB	6.0	9.0	15.6	19.4	25.1	30.9	34.0	32.5	28.2	20.2	13.6	7.2		
		MCWB	2.6	4.0	6.8	8.9	12.2	14.5	16.0	15.5	14.0	10.3	6.5	3.3		
	10%	DB	4.2	7.1	13.5	17.2	22.8	28.9	32.7	31.2	26.4	18.0	11.4	5.1		
		MCWB	1.5	2.9	5.8	8.0	11.5	13.9	15.7	15.3	13.5	9.1	5.5	2.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	7.1	9.8	12.1	15.5	17.6	19.1	19.1	18.0	13.8	9.7	6.7		
		MCDB	8.7	11.5	18.7	23.0	28.0	31.7	29.4	29.9	27.6	21.9	15.6	10.8		
	2%	WB	4.1	5.5	8.6	10.6	13.8	16.1	18.3	17.9	16.3	12.4	8.0	5.1		
		MCDB	6.5	9.9	17.0	20.1	25.0	29.7	29.6	29.0	26.5	20.0	13.9	8.6		
	5%	WB	3.0	4.2	7.3	9.5	12.8	15.2	17.7	17.1	15.3	11.1	6.9	3.8		
		MCDB	5.4	8.1	14.6	18.0	23.1	28.2	29.4	28.7	25.4	18.8	12.6	6.6		
	10%	WB	1.8	3.1	6.3	8.5	11.9	14.5	17.0	16.4	14.3	10.0	5.8	2.6		
		MCDB	4.2	6.4	12.8	16.2	21.3	26.9	29.0	28.3	23.7	17.5	11.0	4.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	7.6	9.5	10.2	11.1	12.9	13.3	12.9	11.9	10.3	8.4	6.5		
		MCDBR	8.0	9.2	12.2	13.7	14.1	15.1	14.7	14.1	13.9	13.0	10.7	7.9		
	5% WB	MCWBR	5.2	5.2	6.3	6.7	6.1	5.8	5.4	5.7	6.1	6.5	6.1	5.0		
		MCWBR	7.5	8.3	11.5	12.6	12.8	13.9	12.7	12.4	11.6	11.7	9.9	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.250	0.262	0.276	0.312	0.315	0.312	0.339	0.334	0.304	0.281	0.261	0.247		
	taud		2.469	2.506	2.554	2.450	2.495	2.541	2.483	2.490	2.563	2.608	2.594	2.516		
	Ebn at Noon		929	967	985	961	960	959	929	926	941	934	914	906		
	Edh at Noon		74	84	90	108	106	101	107	103	89	75	66	65		
(44) All-Sky Solar Radiation	RadAvg		1.99	2.87	4.32	5.53	6.52	7.77	7.53	6.54	5.30	3.65	2.36	1.65		
	RadStd		0.21	0.31	0.29	0.34	0.54	0.55	0.33	0.31	0.36	0.25	0.24	0.21		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page